

Direct Retainers (continue)

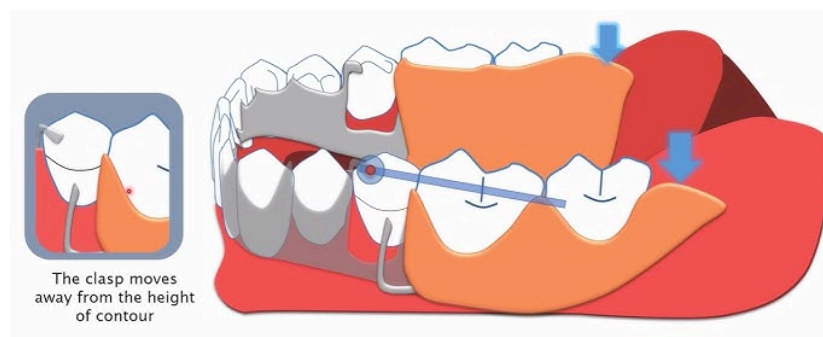
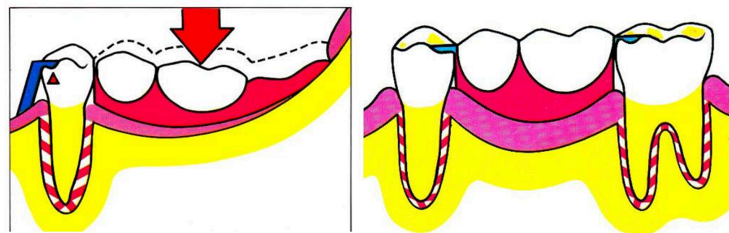
Direct Retainers For Tooth and Tissue-Borne RPDs

It's also known as an infra-bulge or gingival approach clasp because it approaches the undercut areas from the gingival direction.

Tooth- and tissue-borne situations (Class I & II) require special attention in direct retainer selection, due to the stresses created by the rotational movements of the prostheses. When the denture bases are placed under function, rotation occurs about the rest seats of the most posterior abutments. Excessive occlusal forces on the distal-extension portion of the denture could cause a torquing action on the abutment teeth unless direct retainers are designed with **stress-breaking** capabilities.

Stress-releasing clasp assemblies include:

1. The RPI clasp
2. The RPA clasp
3. the combination clasp

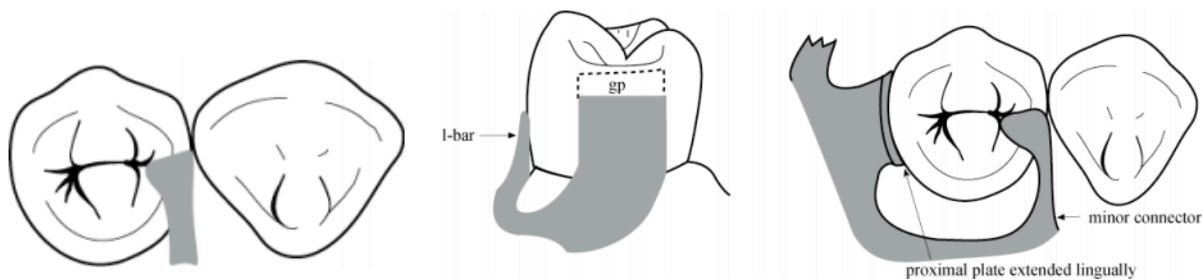


1. The RPI Clasp

The components of this clasp assembly are:

- "R" - rest (always mesial)
- "P" - proximal plate
- "I" - I-bar (retentive arm)

The rest is located on the mesio-occlusal surface of a premolar or mesiolingual surface of a canine. The minor connector is located in the mesio-lingual embrasure but is not in contact with the adjacent tooth (prevents wedging).

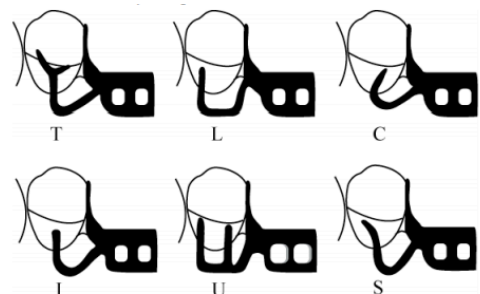


The proximal plate (essentially a wide minor connector) is located on a guide plane on the distal surface of the tooth.

The I-bar clasp is located on the buccal surface of the premolar and the mesio-buccal surface of the canine.

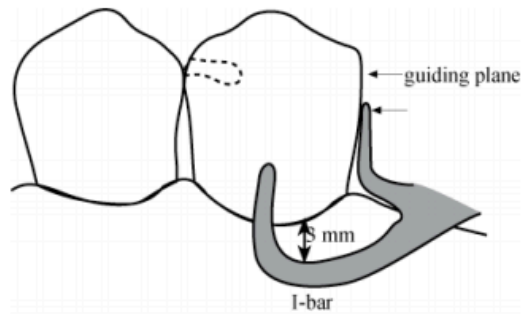
- The bar clasp is a cast clasp that arises from the partial denture framework and approaches the retentive undercut from the gingival direction.
- Retentive clasps are identified by the shape of the retentive terminal, i.e. T, Y, L, I, U, and S.

The shape is unimportant as long as the direct retainer is mechanically and functionally stable, covers minimal tooth structure with minimum display (the I bar most often meets these requirements)



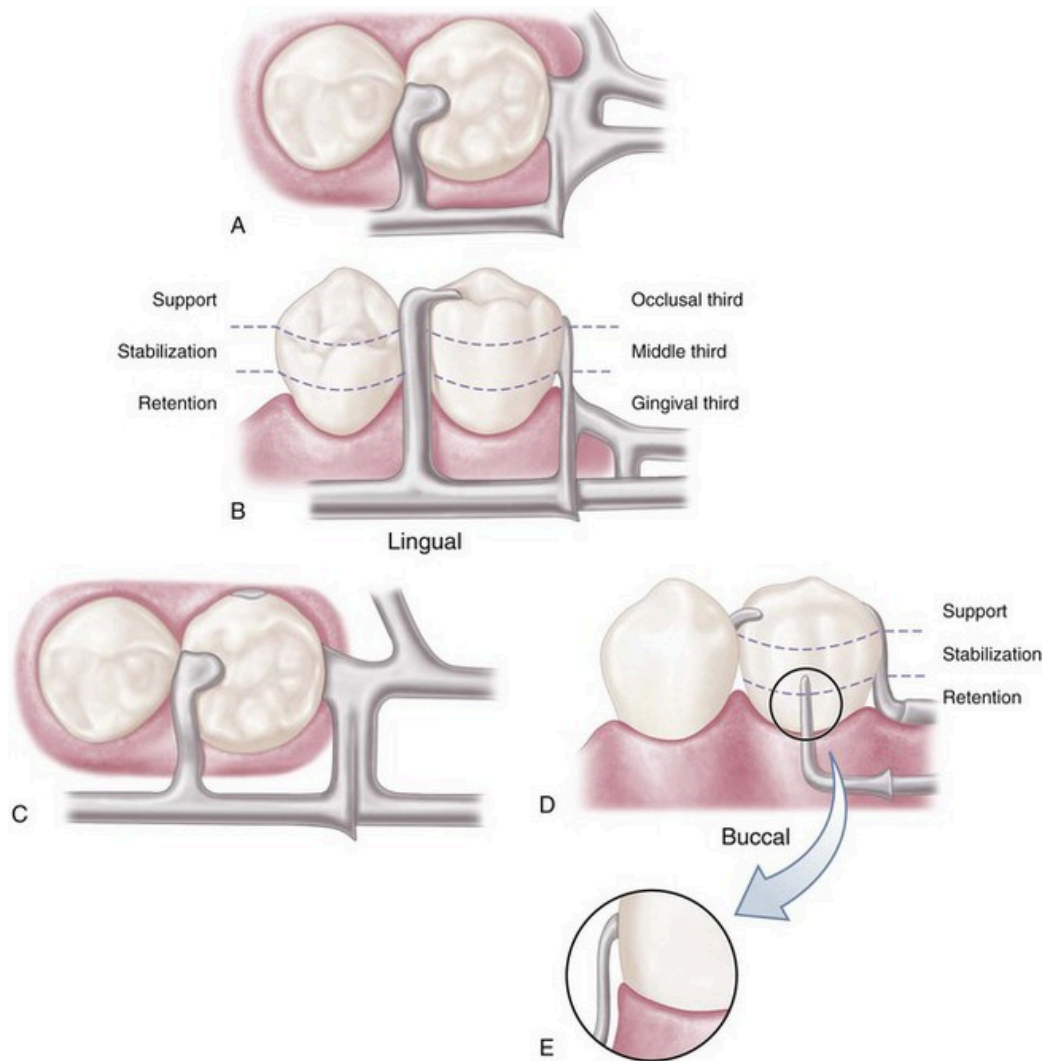
- I-bar should be located in the gingival third of the buccal or labial surface of the abutment in a 0.01-inch (≈ 0.25 mm) undercut.

d. Soft tissue relief is provided under the approach arm with wax to prevent tissue impingement.



Contraindications to the RPI Clasp

- 1) Insufficient depth of the vestibule. (The bend in the I-bar should be located at least 3 mm from the gingival margin.)
- 2) pronounced frenal attachments in the area – impingement
- 3) deep cervical undercuts - food entrapment



A: Occlusal view. Parts (proximal plate minor connector, rest with minor connector, and retentive arm) tripod the abutment to prevent its migration. **B:** The proximal plate minor connector extends just far enough lingually so that it combines with the mesial minor connector to prevent the lingual migration of the abutment. **C:** On narrow or tapered abutments (mandibular first premolars), the proximal plate should be designed to be as narrow as possible but still sufficiently wide to prevent lingual migration. **D:** I-bar retainer located at the greatest prominence of the tooth in the gingival third. **E:** Mesial view of the I-bar illustrating the retentive tip relationship to the undercut.

2. RPA Clasp

R: Rest

P: Proximal plate

A: Akers clasp

Shape: This clasp assembly is similar to the RPI design, except a wrought wire circumferential clasp (Akers) is used instead of the I-bar.

Position: This clasp arises from the proximal plate and terminates in the mesiobuccal undercut.

Indication: It is used when there is insufficient vestibule depth or when a severe tissue undercut exists.



3. Combination Clasp

Shape: The combination clasp is similar to the cast circumferential clasp, with the exception that the retentive arm is fabricated from a round wrought wire.

Another strategy to reduce the effect of the Class I lever in distal extension situations is to use a flexible component in the "resistance arm," which is the strategy employed in the combination clasp. The combination clasp consists of a wrought-wire retentive clasp arm (round, uniformly tapered 18-gauge platinum-gold-palladium alloy or chrome-cobalt alloy wrought-wire) and a cast reciprocal clasp arm.



- It can be placed in a 0.02-inch ($\approx 0.5\text{mm}$) undercut due to its flexibility (allows lower placement for better esthetics).
- It can be used in tooth-borne cases.

Advantages:

- o high flexibility.
- o high adjustability.
- o better aesthetic appearance of the wrought-wire retentive arm than circumferential clasp arms.

Disadvantages:

- o It involves additional fabrication steps, particularly when high-fusing chromium alloys are used.
- o It may be distorted by the patient's careless handling.



RPI



RPA



combination clasp

4. Infrabulge clasp:

It is designed so that the bar arm arises from the border of the denture base, either as an extension of a cast base or attached to the border of a resin base. It is made more flexible than the usual bar clasp arm.

Advantages:

- a. Its interproximal location may be used for aesthetic advantage.
- b. Increased retention without tipping action on the abutment.
- c. There is less chance of accidental distortion from its proximity to the denture border.



Lingual aspect open

Lingual



lingual aspect plated



Buccal

(Clasp arm arises from border of metal base.)



(Clasp arm is attached to buccal flange of resin denture base with auto-polymerizing resin.)